

Real-World Data Sources

A Comprehensive Guide to Connection Types, Ingestion Pipelines, and Transformation Models
in Power BI

Module 1: Foundations

What defines a data source and how does BI manage storage connections?

What is a Data Source?

Core Concept

A data source is simply any repository where digital information is systematically stored.

In modern ecosystems, this ranges from simple flat files on a local drive to complex multi-server transactional relational databases hosted in the cloud.

The Water Analogy

Think of your data sources as water tanks scattered across a massive landscape.

Without an ingestion system, they remain isolated. **Power BI acts as the pipe network**, drawing all these disjointed streams together into a single, high-flow tap.

Flat Files: Excel & CSV

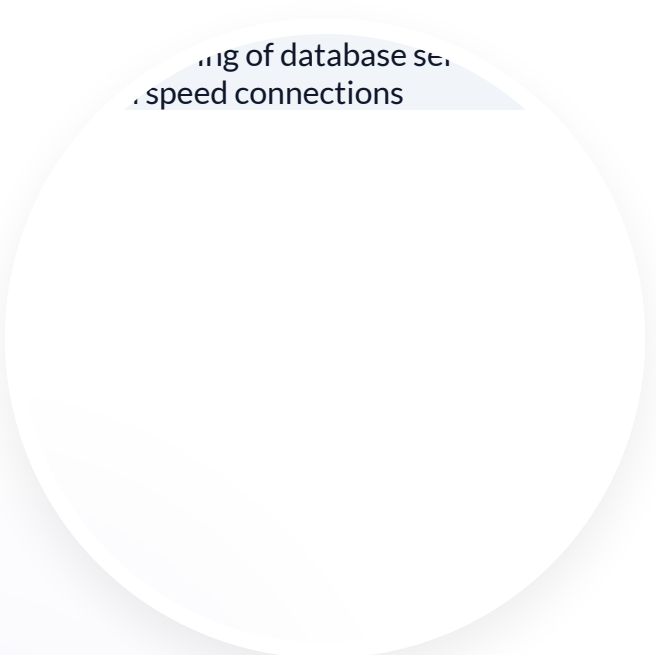
The Universal Ingestion Standard

Excel Files (.xlsx): Best suited for formatted, multi-tab business spreadsheets. Excel provides local structure, layout, and manually computed formulas.

CSV Files (.csv): Lightweight, comma-separated plain text. CSV drops all native formatting to prioritize high-speed, raw-data transfers across networks.

"Excel and CSV serve as the foundational, entry-level endpoints for almost all initial BI development."

Relational Databases: SQL



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speed connections

SQL Server Architectures

When organizations outgrow flat files, they consolidate business-critical records in relational databases like SQL Server (e.g., enterprise employee or transaction systems).

The Warehouse Analogy: SQL Server functions like an immense, ultra-modern logistics warehouse. Everything is perfectly cataloged, stored in designated slots, and optimized for rapid security-managed extraction.

Web & API Integration



HTML Web Tables

Power BI reads native HTML tables from static pages. Useful for appending public datasets, such as active population metrics directly from Wikipedia pages.



Live APIs

Application Programming Interfaces provide automated live feeds. Perfect for external parameters like weather trends, stock tickers, or mapping services.



JSON Awareness

Modern APIs transfer data inside lightweight, nested JSON structures. Power BI's Query Engine natively parses and flattens these arrays into standard columns.

Batch Folder Loading

100%

AUTOMATED APPEND

Consolidating Multi-File Data

Organizations often generate periodic reporting files separately (e.g., *Jan.xlsx*, *Feb.xlsx*, *Mar.xlsx*).

By establishing a connection directly to a designated **Folder directory** rather than to individual flat files, Power BI will dynamically grab, format, and aggregate all structural files in a single ingestion sweep.

Ingestion: Import vs DirectQuery

In-Memory Import

Data is completely extracted from the source and stored in compressed formats within the Power BI database.

- Blazing-fast visual render performance.
- Access to the full suite of DAX calculation functions.
- Requires scheduled background refreshes to stay current.

Live DirectQuery

Power BI retains no cached copy of the data. It sends live database queries directly to the source system as users click visuals.

- Real-time, instant data updates.
- Ideal for massive data volumes (terabytes+).
- Visual speed is restricted by original database performance.

The Connection Analogy

Photo Snapshot vs. Live Stream

Import Mode functions exactly like taking a high-definition photograph. It captures a crisp, pristine copy of the landscape at that precise moment. If the underlying data updates, you must take a new photo (refresh).

DirectQuery is like watching a live security CCTV camera stream. It displays what is happening in the source system in real-time, although network quality may occasionally impact resolution and response speed.

Load vs. Transform Data

-  **Direct Loading:** Pushes original datasets immediately into your reporting model. Recommended only for perfectly clean, standardized warehouses.
-  **Power Query Transformation:** Opens the ETL editor to clean, restructure, and transform files prior to modeling.
-  **Essential Data Sanitization:** Allows developers to instantly strip duplicate rows, re-assign column datatypes, filter empty records, and split compound fields.

 Abstract concept representing clean data pipelines, filtering, and computer processes

Discussion & Questions

Let's test our collective understanding on the key architectural decisions:

1. Which source do established companies rely on most frequently?
2. Why would real-time (DirectQuery) connections be critical for monitoring systems?
3. What happens to visual performance if our data is loaded without transforming messy rows?